Multi-Modal Periodic Energy-Based Navigation System for Autonomous Drone Homing

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***Abstract*—Autonomous drone homing remains challenging when Global Navigation Satellite System (GNSS) signals are unavailable, obstructed, jammed, or unreliable. This paper presents a multi-modal periodic energy-based navigation system that combines optical, thermal, and acoustic sensing in a staged GPS-independent homing architecture. A designated base station periodically emits identifiable signals in multiple physical modalities. During the long-range phase, optical and thermal measurements provide coarse target direction; during the near-field phase, a microphone array provides acoustic localization using Time Difference of Arrival (TDOA), Angle of Arrival (AOA), and signal-strength information. The architecture integrates signal acquisition, filtering, feature extraction, sensor fusion, localization, navigation, flight control, telemetry, and safety monitoring in a closed loop. A web-based ground-control layer supports live telemetry, mission replay, event logging, and health monitoring. The prototype and development workflow are evaluated using signal-processing, localization, integration, telemetry, and replay-oriented metrics. The reported numerical values are development/replay observations rather than final free-flight measurements. The work provides a practical architecture for future hardware-in-the-loop and controlled flight validation in GPS-denied environments.**

***Index Terms*—Autonomous Drone, GPS-Free Navigation, Drone Homing, Sensor Fusion, Optical Sensing, Thermal Sensing, Acoustic Localization, TDOA, AOA, Periodic Beacon, UAV Navigation.**

I. Introduction

Unmanned Aerial Vehicles (UAVs) are increasingly used for inspection, surveillance, logistics, environmental monitoring, disaster response, and autonomous exploration. Reliable localization is fundamental to these applications because the navigation stack must continuously estimate the vehicle state and generate safe motion commands [1]–[4]. Conventional outdoor autonomous flight frequently relies on GNSS/GPS for global position and return-to-home functions. However, satellite-based positioning can degrade or become unavailable indoors, underground, beneath structures, in dense urban canyons, or under intentional interference [5]–[7].

Vision-based navigation and visual-inertial odometry can provide rich geometric information, but feature extraction and tracking may degrade under low light, smoke, dust, blur, occlusion, repetitive texture, or other visual degradation [8]–[11]. LiDAR and camera fusion can improve geometric robustness, but additional sensing hardware increases payload, computation, calibration effort, and system complexity [12]. RF and ultra-wideband (UWB) systems can provide absolute or relative positioning when infrastructure is available, but their performance can be affected by multipath, propagation conditions, anchor geometry, and interference [13]–[16]. Acoustic localization is attractive for short-range directional estimation and can use microphone arrays with TDOA/GCC-based processing, but ambient noise, reverberation, reflections, and array geometry affect accuracy [17]–[20].

Thermal sensing provides complementary information in conditions where visible imagery is weak or illumination is poor, and UAV research has explored thermal imagery for localization and perception in challenging environments [21]–[23]. The broader GNSS-denied navigation literature increasingly emphasizes sensor fusion because no single modality is robust under all environmental conditions [5], [24]. Kalman filtering and probabilistic state estimation are widely used to combine measurements with different uncertainty, rates, and failure modes [25]–[28].

These observations motivate a staged multi-modal strategy rather than a single estimator operating identically throughout the mission. The proposed system uses a designated periodic beacon as an explicit homing reference. Optical and thermal measurements are used for long-range target acquisition, while acoustic TDOA/AOA information is activated for near-field refinement. The design therefore assigns each modality to the region where it is most useful, while a fusion layer controls

the contribution of each measurement according to confidence and signal quality. The architecture also couples localization to navigation state, flight control, telemetry, and safety rather than treating localization as an isolated module [1], [29], [30].

A. Research Gap

Existing GPS-denied UAV solutions commonly optimize a dominant modality such as vision, inertial sensing, LiDAR, UWB, or acoustics. Such systems can perform well under their intended conditions, but their failure modes differ. Vision can be weakened by illumination and scene degradation; RF/UWB systems require appropriate infrastructure and are sensitive to propagation conditions; acoustic systems are most useful at shorter ranges and can be affected by reverberation; and inertial estimation accumulates drift without external correction [3], [6], [8], [11], [13], [17].

The research gap addressed by this work is therefore the lack of a compact, staged homing architecture that deliberately combines periodic optical, thermal, and acoustic energy cues, switches modalities according to the estimated homing phase, and connects the resulting localization to a complete navigation, control, telemetry, and safety loop. The contribution is not the invention of any individual sensing technique; rather, it is the system-level integration and staged use of complementary physical modalities for a designated GPS-independent homing task.

II. Related Work

GPS-based UAV navigation provides reliable global positioning and is widely used for autonomous flight and return-to-home functions. However, its dependence on satellite visibility limits operation in indoor, obstructed, underground, and intentionally denied environments [6], [31], [32]. Vision-based navigation and visual-inertial systems provide rich environmental information and can support localization without GNSS, but their performance depends on illumination, texture, feature availability, computational resources, and scene conditions [3], [8], [9], [11].

RF and UWB approaches provide useful relative or absolute localization when suitable infrastructure is available. Their accuracy can nevertheless depend on anchor geometry, multipath, propagation conditions, and interference [13], [15], [16]. Acoustic localization provides another alternative, particularly for directional and short-range localization. Microphone arrays combined with TDOA, GCC, and AOA processing can estimate source direction, although noise, reverberation, reflections, and microphone geometry influence accuracy [17]–[20].

Multi-sensor fusion provides a means of combining complementary measurements and reducing dependence on a single sensing modality [25], [27], [29]. The proposed work extends this principle to a designated periodic homing reference by assigning optical and thermal sensing to long-range target acquisition and acoustic localization to near-field refinement. This staged organization connects sensing, localization, navigation, control, telemetry, and safety within one GPS-independent homing workflow.

III. Problem Statement and Objectives

A. Problem Statement

The problem considered in this work is the autonomous return of a UAV toward a designated base station when GNSS/GPS cannot be assumed to be continuously available. The drone must detect the base-station beacon, estimate its relative direction and/or position, select an appropriate navigation mode, generate flight-control commands, and continue the measurement–decision–motion loop until target alignment is achieved. The system must also tolerate changes in signal quality and provide a safe response to localization uncertainty, communication loss, or sensor failure.

The design is intentionally reference-based: the base station provides a known periodic source rather than requiring a prebuilt global map. This makes the problem suitable for indoor recovery, structured GPS-denied areas, controlled search-and-rescue scenarios, and other deployments where a homing reference can be installed [5], [6], [33].

B. Objectives

The primary objective is to develop a GPS-independent UAV homing architecture that combines optical, thermal, and acoustic sensing with a periodic beacon. The system is designed to use optical and thermal measurements for long-range target acquisition, followed by microphone-array TDOA/AOA processing for near-field directional refinement. A confidence-aware fusion mechanism is intended to combine heterogeneous observations according to signal quality and uncertainty. The navigation layer connects localization estimates to navigation states and PID-based flight-control commands, while the telemetry and ground-control layer provides mission monitoring, replay, event logging, and safety supervision. The complete workflow is evaluated using signal, localization, integration, telemetry, and homing-performance metrics while clearly distinguishing development and replay observations from future physical-flight measurements.

IV. Proposed Methodology

The proposed methodology is organized as a closed-loop pipeline:

Base Station → Sensors → Signal Processing → Sensor Fusion → Localization → Navigation → Flight Control → Actuation.

A. System Architecture and Overall Flow

The proposed system follows a layered architecture consisting of a base-station layer, onboard sensor layer, signal-processing layer, localization and fusion layer, navigation and decision layer, flight-control layer, telemetry layer, and safety layer. This separation permits individual modules to be tested independently before complete system integration. The architecture follows established UAV and robotics system-modeling principles [34]–[36] while adapting the localization

source to a dedicated multi-modal homing beacon [1], [30], [31], [34].

The overall operation begins at the base station, which periodically generates identifiable energy cues. The drone sensor layer acquires these signals and passes them to the signal-processing stage. Filtering, frequency analysis, feature extraction, and noise suppression are then performed independently for each sensing modality. The resulting features are passed to the fusion layer, where confidence and signal-quality information determine the contribution of each modality.

The localization layer converts the fused measurements into target direction or relative position. The navigation layer interprets this estimate according to the current homing state and generates the required movement direction. The flight-control layer converts these navigation commands into attitude and throttle corrections, which are finally executed through the flight controller, ESCs, and motors. New sensor measurements are continuously acquired after actuation, thereby closing the feedback loop.

B. Base Station and Periodic Energy Beacon

The base station acts as the explicit homing reference. It periodically emits three identifiable signal classes: optical, thermal, and acoustic. The periodic structure provides temporal information that helps distinguish the intended beacon from unrelated environmental signals. Beacon parameters can include carrier or modulation frequency, pulse period, intensity, duty cycle, and transmission state.

Periodic beaconing is particularly useful in a reference-based architecture because detection becomes a hypothesis-testing problem rather than an open-ended search over the entire environment. The signal-processing layer can correlate observations with expected temporal and spectral patterns before passing them to the localization stage [18], [37], [38].

C. Drone Sensor Layer

The drone contains sensing elements matched to the three transmitted modalities. The optical or infrared channel measures beacon intensity and directional evidence. The thermal channel detects the thermal signature and provides complementary information when visible-light conditions are unfavorable. The acoustic channel uses a spatially separated microphone array so that the same beacon event is observed at different sensor locations.

Microphone-array geometry is important because TDOA is converted into spatial constraints through the known sensor separation and propagation velocity [17], [39], [40]. Sensor timestamps and sampling rates must remain sufficiently synchronized for the expected localization accuracy.

D. Signal Processing and Feature Extraction

Raw observations can contain environmental noise, multipath components, reverberation, sensor noise, and irrelevant frequency content. The signal processing stage therefore performs filtering, noise reduction, frequency analysis, peak detection, and feature extraction.

For an acoustic signal $x(t)$, the continuous Fourier transform is

$$X(f) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi f t}\, dt. \tag{1}$$

The dominant frequency and spectral energy are compared with the expected beacon signature. Classical digital signal-processing formulations provide the frequency-domain and filtering foundations used in this stage [41]–[44].

A basic signal-to-noise ratio (SNR) measure is

$$SNR = 10 \log_{10} \left(\frac{P_s}{P_n} \right), \tag{2}$$

where P_s and P_n are signal and noise power, respectively. For acoustic localization, generalized cross-correlation and GCC-PHAT are appropriate tools for estimating inter-channel delay under noise and reverberation [18]–[20].

E. Long-Range Optical and Thermal Homing

During long-range navigation, optical and thermal observations provide coarse direction toward the base station. The system does not require centimeter-level accuracy at this stage; it requires a sufficiently reliable homing direction. The navigation layer evaluates detection confidence, signal strength, and consistency over successive observations.

The use of two complementary long-range modalities is intended to reduce the probability that one environmental condition completely disables target acquisition. Thermal sensing can provide useful information in low-light or nighttime conditions, while optical sensing can provide stronger directional cues when the beacon is visually distinct [21], [23].

F. Near-Field Acoustic Localization

When the estimated range reaches a predefined transition region, acoustic localization is activated. For microphones i and j, the TDOA is

$$\Delta t_{ij} = t_i - t_j. \tag{3}$$

The corresponding path-length difference is

$$\Delta d_{ij} = c \Delta t_{ij}, \tag{4}$$

where c is the speed of sound. Multiple microphone-pair measurements produce multiple geometric constraints on the source direction. GCC-PHAT, SRP-PHAT, AOA estimation, and hyperbolic localization can be used as progressively more advanced implementations [17]–[20], [45]–[48].

G. Sensor Fusion and Confidence Management

The localization layer combines optical, thermal, and acoustic observations. A generalized confidence-weighted estimate is expressed as

$$\hat{z} = w_o z_o + w_t z_t + w_a z_a, \tag{5}$$

where z_o, z_t, and z_a denote modality-specific measurements and the weights satisfy an appropriate normalization. The

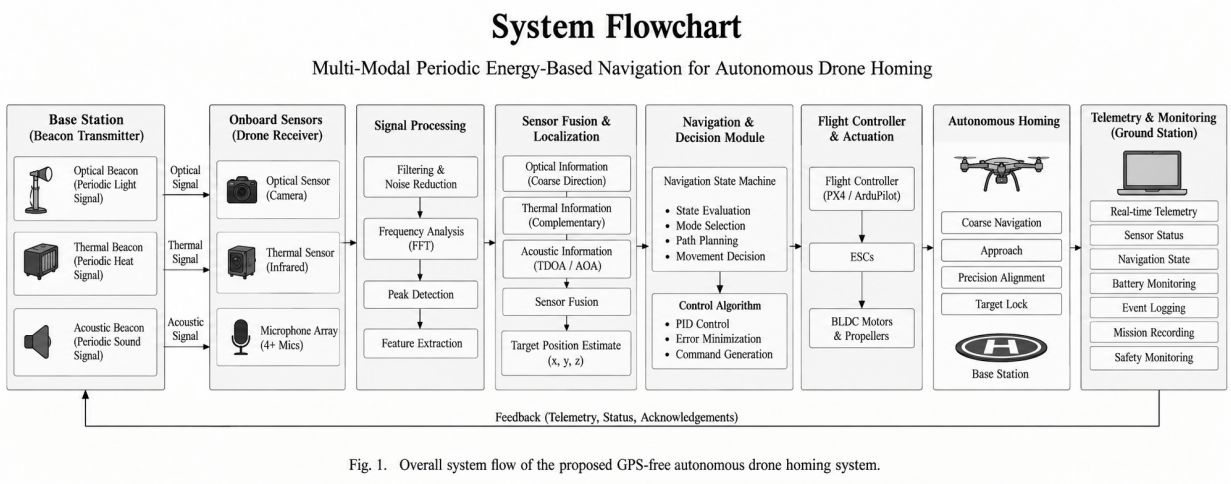


Fig. 1. Overall system flow of the proposed GPS-free autonomous drone homing system.

Fig. 1. Overall system flow of the proposed GPS-free autonomous drone homing system.

weights are adjusted using signal quality, noise level, detection confidence, and navigation mode.

For a practical implementation, the weights may be interpreted as normalized inverse uncertainty or confidence scores. A Kalman or extended Kalman filter can subsequently be introduced when a dynamic state model and covariance estimates are available [25]–[27], [29], [49]–[54].

H. Multi-Modal Homing Methodology

The complete homing procedure follows a staged strategy. Initially, the drone searches for the periodic beacon and confirms the target using optical and thermal observations. Once sufficient confidence is obtained, the system enters coarse homing and progressively reduces the relative distance to the base station. When the estimated range enters the predefined near-field region, acoustic localization is activated. TDOA and AOA measurements then provide additional directional information for final approach and alignment. The process terminates when the target-lock condition satisfies the required position and confidence thresholds.

I. Navigation State Machine and Control

The navigation state machine contains system initialization, base-station detection, optical acquisition, thermal confirmation, long-range localization, coarse homing, near-field transition, acoustic localization, precision alignment, target lock, and homing completion. Transitions depend on sensor availability, signal quality, estimated distance, and localization confidence.

The controller generates desired heading, pitch, roll, and throttle corrections. A PID controller is represented by

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau)d\tau + K_d \frac{de(t)}{dt}, \qquad (6)$$

where $e(t)$ is the navigation error. PID and geometric control are established approaches for multirotor stabilization and tracking [1], [30], [55]–[60].

J. System Architecture Details

The major functional layers and their roles are summarized in Table I. The base-station layer generates the reference signals, while the onboard sensing layer performs simultaneous acquisition of the available physical modalities. The signal-processing layer converts raw measurements into usable features and removes unwanted environmental components. Localization and fusion then estimate the target state using modality-specific measurements and confidence information.

The navigation layer determines the appropriate homing state and converts the localization estimate into movement objectives. The flight-control layer is responsible for stabilization and actuator timing, while the telemetry layer provides mission observability. Finally, the safety layer monitors critical system conditions and prevents the navigation loop from continuing under unsafe or unreliable conditions.

The onboard processing platform may be an ESP32-class controller for lightweight signal acquisition or a Raspberry Pi-class computer when greater processing capacity is required. The flight controller remains responsible for low-level stabilization and actuator timing.

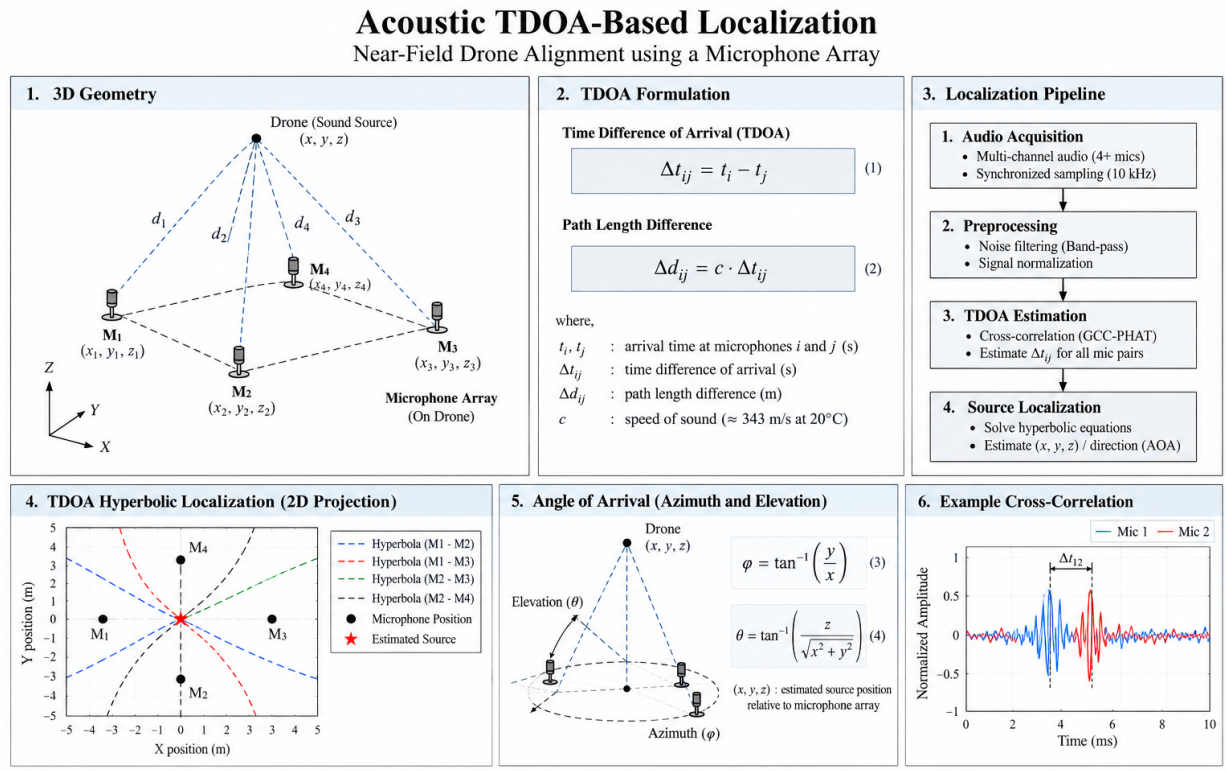


Fig. 5. Acoustic TDOA-based localization using a microphone array for near-field drone alignment.

Fig. 2. Acoustic TDOA-based localization using a microphone array for near-field drone alignment.

TABLE I
MAJOR SYSTEM COMPONENTS

Layer	Function
Base Station	Periodic optical, thermal, and acoustic beacon generation
Sensor Acquisition	Collection of optical, thermal, and acoustic measurements
Signal Processing	Filtering, FFT, noise reduction and feature extraction
Localization	AOA, TDOA and signal-strength based estimation
Sensor Fusion	Combination of multi-modal measurements
Navigation	Homing state selection and movement decision
Flight Control	PID/control commands for drone movement
Actuation	ESC, BLDC motor and propeller control
Telemetry	Monitoring of flight and sensor parameters
Safety	Detection of faults, signal loss and critical conditions

K. Ground Control and Telemetry

The ground-control station provides a unified monitoring layer for mission state, sensor status, localization, flight telemetry, communication health, battery state, and historical mission data. The displayed information includes drone position and target direction, optical, thermal, and acoustic signal strength, TDOA and AOA measurements, localization confidence, battery state, flight attitude, motor telemetry, communication latency, packet loss, navigation state, and fault logs.

A centralized telemetry provider allows the visualization layer to consume live hardware data, replayed missions, or development streams without changing the user interface. Mission history can contain identifiers, timestamps, navigation states, localization measurements, signal information, communication events, and system-health information. Telemetry replay is useful for deterministic software validation before free-flight testing [31], [61], [62].

L. Security, Privacy and Safety

The ground-control interface should implement authenticated access and role-based authorization. Sensitive operations such as ARM, DISARM, START, and ABORT should require appropriate authorization. Communications between the drone, base station, and ground station should use authenticated and encrypted channels where supported. Security controls should also protect telemetry and mission history from unauthorized modification [63], [64].

Safety monitoring continuously evaluates battery state, sensor availability, communication quality, localization confi-

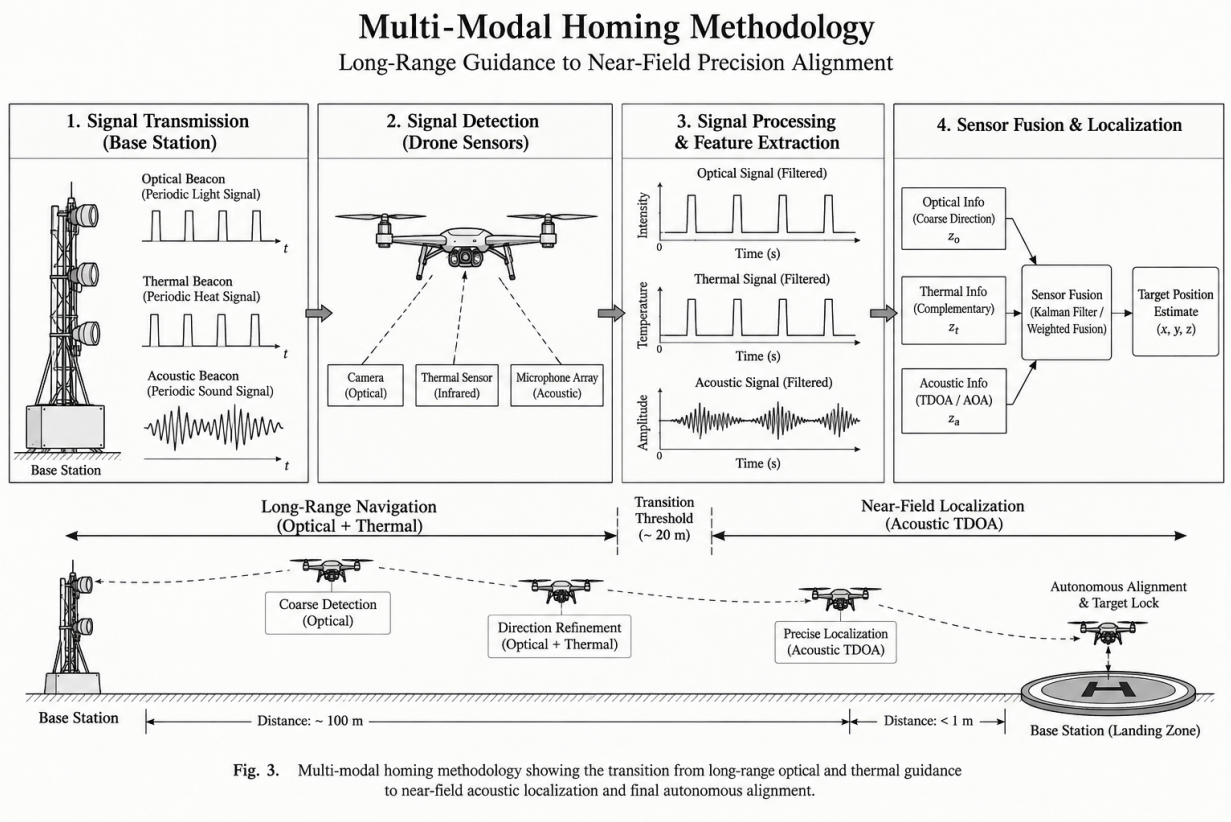


Fig. 3. Multi-modal homing methodology showing the transition from long-range optical and thermal guidance to near-field acoustic localization and final autonomous alignment.

Fig. 3. Multi-modal homing methodology showing long-range optical and thermal guidance followed by near-field acoustic localization and final autonomous alignment.

dence, and critical faults. If communication is lost or localization becomes unreliable, the system should enter a predefined safe state rather than continuing uncontrolled navigation. The distinction is important because standard autopilot return-to-home modes may themselves require a valid global position estimate; therefore a GPS-free homing architecture needs its own explicit fallback logic [32], [65].

V. Implementation and Technology Stack

The prototype uses an embedded processing and flight-control stack together with a web-oriented telemetry layer. The proposed technology stack is shown in Table II.

TABLE II
Proposed Technology Stack

Category	Technology
Embedded Processing	ESP32 / Raspberry Pi
Programming	C/C++, Python
Signal Processing	MATLAB, Python, FFT, digital filtering
Localization	AOA, TDOA, signal-strength analysis
Control	PID / adaptive control
Flight Hardware	Flight Controller, ESC, BLDC Motors
Sensors	Optical/IR, Thermal, Microphone Array
Communication	Wi-Fi / LoRa / Serial
Ground Station	React / Next.js, TypeScript
Visualization	Recharts / custom telemetry plots
Data	CSV, JSON, KML
Development	VS Code, Git, GitHub
Security Testing	Wireshark, OWASP ZAP, Postman

The software architecture separates data acquisition from visualization. A telemetry provider exposes a common interface for live hardware data, recorded mission data, and development data, allowing the same ground-control interface to be used throughout the development lifecycle.

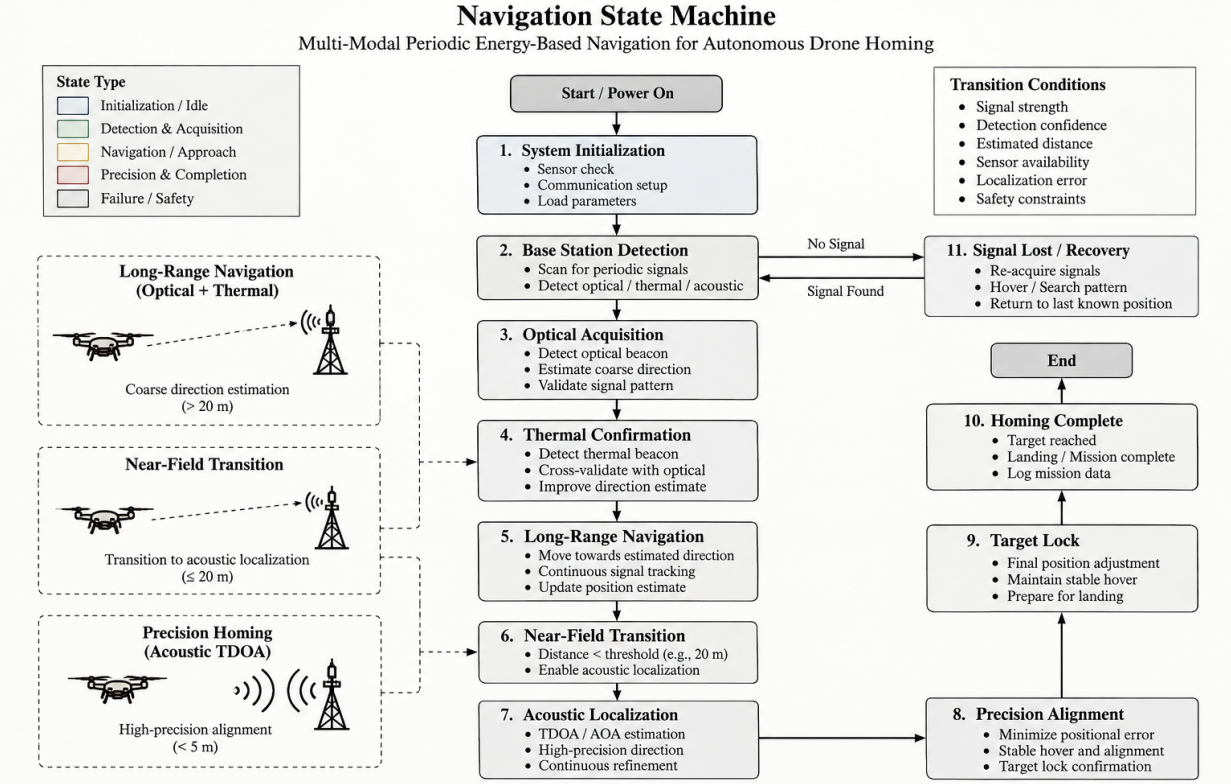


Fig. 4. Navigation state machine showing the transition from beacon acquisition to autonomous homing and target lock.

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VI. Experimental Results and Analysis

The evaluation is divided into signal-level, localization-level, integration-level, telemetry-level, and homing-level observations. The current paper reports development and replay observations from the implemented software workflow and prototype integration. These values must not be interpreted as statistically complete free-flight validation; controlled flight experiments remain a required next step.

A. Evaluation Metrics

The principal evaluation metrics are localization error, signal-to-noise ratio, detection confidence, fused confidence, homing success rate, navigation update rate, telemetry rate, latency, and packet loss.

Localization error is defined as the Euclidean distance between the reference and estimated target positions:

$$E = \sqrt{(x - \hat{x})^2 + (y - \hat{y})^2 + (z - \hat{z})^2}. \tag{7}$$

SNR evaluates the ratio of detected beacon power to noise power. Detection confidence represents the confidence assigned to the detected beacon after modality-specific processing, while fused confidence represents the confidence after combining the active modality estimates. Homing success rate represents the percentage of replay scenarios reaching the defined target-alignment condition. Navigation update rate measures the number of localization and navigation decisions generated per second. Telemetry rate, latency, and packet loss characterize communication and monitoring performance.

B. Development and Replay Results

Table III retains the observations from the current development stage. They are presented as engineering observations rather than claims of final field performance.

TABLE III
Development and Replay Observations

Parameter	Observation
Initial Homing Range	100 m
Near-Field Transition	20 m
Final Alignment	0.8 m
Navigation Update Rate	10 Hz
Fused Confidence	92.4%
Average Localization Error	1.5 m
Homing Success in Replay Scenarios	≈90%
Telemetry Rate	10–20 Hz

The telemetry sequence demonstrates a staged reduction in relative distance. During the initial phase, optical and thermal measurements support coarse homing. As the estimated distance approaches the near-field threshold, the acoustic stage is activated. The TDOA stage then contributes directional information for final alignment. This behavior is consistent

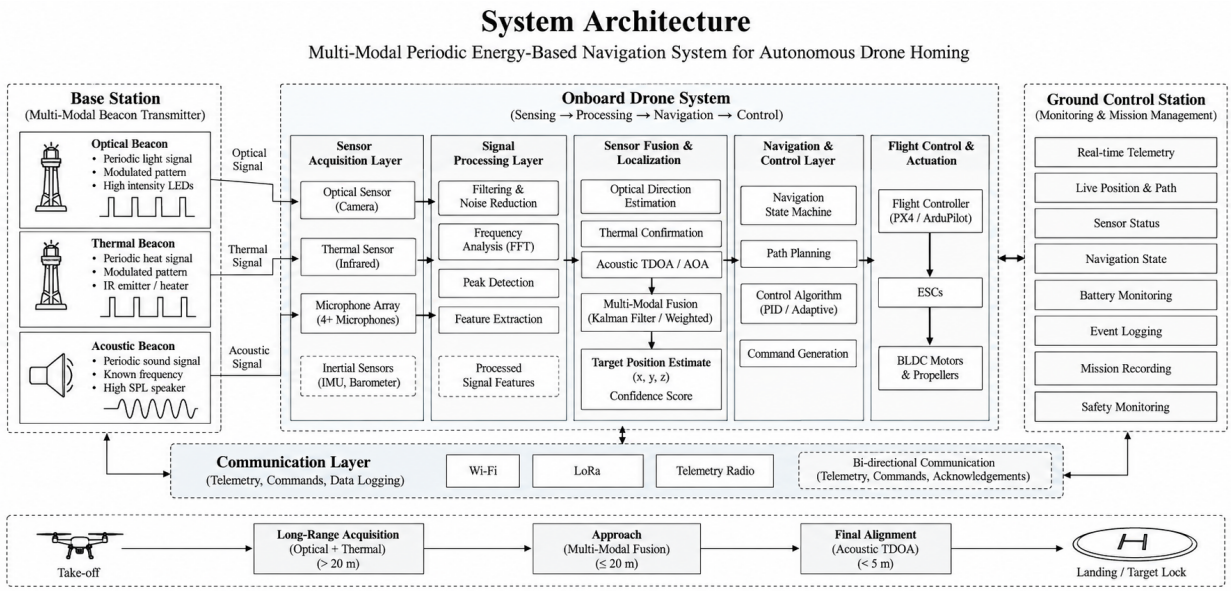


Fig. 2. Layered architecture of the multi-modal autonomous drone homing system.

Fig. 5. Layered architecture of the multi-modal autonomous drone homing system.

with the intended architecture in which different modalities dominate at different navigation stages.

C. Signal and Localization Analysis

Signal-processing performance is assessed using frequency detection, SNR, peak detection, and filtering behavior. Known target positions can be used to calculate AOA/TDOA localization error. Future controlled experiments should evaluate the same metric over repeated trials under different noise, lighting, temperature, distance, and reverberation conditions.

The current average localization error of 1.5 m is therefore a development/replay observation, not a certified flight-test accuracy. Likewise, the 92.4% fused confidence represents the internal confidence measure used by the current workflow and should not be equated directly with probability of true position correctness.

D. Integration and Telemetry Analysis

The complete chain is evaluated as

$$\text{Sensor Data} \rightarrow \text{Processing} \rightarrow \text{Localization} \rightarrow \text{Navigation} \rightarrow \text{Control}.$$

Communication testing considers packet rate, latency, packet loss, and reconnection behavior. The telemetry architecture also maintains temporal consistency between distance, signal strength, localization confidence, navigation state, and event history. This is important because independently generated values can produce misleading demonstrations even when each value is numerically plausible.

E. Prototype Observation

The implemented UAV prototype and sensor assembly shown in Figs. 6 and 7 demonstrate the physical integration direction of the project. The prototype should be treated as a hardware-integration milestone rather than evidence that the full multi-modal flight loop has already been validated under all environmental conditions.

VII. Comparative Analysis

To position the proposed system against representative GPS-denied or homing approaches, Table IV provides a qualitative architecture-level comparison. The comparison is intentionally based on design characteristics reported in the literature rather than claiming identical benchmark conditions.

GNSS return-to-home is mature and widely supported by autopilot stacks, but standard return modes can require a valid global position estimate [31], [32]. Vision/VIO/SLAM avoids GNSS dependence and can provide rich local geometry, but its robustness is linked to visual observability and computational resources [8], [9], [11]. UWB and related RF localization can provide high-accuracy local positioning, but anchor configuration and propagation conditions become important system constraints [13], [16]. The proposed architecture instead trades the universality of map-based navigation for a deliberately installed homing reference and complementary energy modalities.

The principal advantage of the proposed approach is therefore architectural: it does not require every modality to solve the complete navigation problem. Optical and thermal sensing handle target acquisition, while acoustic sensing is reserved for a near-field precision role. This staged allocation can reduce unnecessary computation and provides explicit failure points for monitoring. The trade-off is the requirement for a designated beacon and careful sensor calibration.

TABLE IV
QUALITATIVE COMPARISON OF REPRESENTATIVE UAV NAVIGATION AND HOMING APPROACHES

Criterion	**GPS/GNSS-based RTH**	**Vision/VIO/SLAM**	**RF/UWB Localization**	**Proposed Multi-Modal Homing**
Primary Reference	Satellite/global position	Scene features and motion	Anchors/beacons	Dedicated periodic multi-modal beacon
GNSS Dependence	High	None during local operation	None	None
Infrastructure	Satellite visibility	Camera/environment	Anchors/infrastructure generally required	One designated base station
Long-Range Target Acquisition	Strong when GNSS is valid	Environment dependent	Strong within radio coverage	Optical + thermal coarse guidance
Near-Field Directional Refinement	Limited by GNSS resolution/use case	Strong if visual features remain trackable	Strong with suitable geometry	Acoustic TDOA/AOA refinement
Low-Light Robustness	Independent of illumination	Potentially reduced	Independent of illumination	Thermal channel provides complementary cue
RF Interference Sensitivity	Vulnerable to GNSS interference/jamming	Low	Potentially significant	Low for the core homing cues
Acoustic Noise Sensitivity	Low	Low	Low	Present in acoustic near-field stage
Drift Tendency	Low when GNSS is healthy	Can accumulate without global correction	Low to moderate depending on estimator	Bounded by external beacon observations
Main Limitation	GNSS denial/obstruction	Illumination, texture, compute	Propagation, anchors, multipath	Calibration, beacon dependency, environmental noise
System-Level Contribution	Global navigation	Local perception/localization	Absolute/relative localization	Staged sensing + fusion + homing + telemetry

VIII. LIMITATIONS

The proposed system has several limitations. First, acoustic localization can be affected by environmental noise, reflections, and reverberation [17], [19]. Second, optical detection can be affected by visibility and illumination, while thermal detection can be influenced by ambient temperature, thermal clutter, and competing heat sources [21], [23].

The system currently depends on a designated beacon source. The base station therefore has to be positioned and configured for the intended environment. Unlike map-based VIO/SLAM approaches, the method does not attempt to infer a complete global map, so it is not a universal replacement for general-purpose navigation [11], [28].

Experimental calibration is also necessary. Sensor alignment, microphone-array geometry, beacon characteristics, filtering parameters, confidence thresholds, and controller gains can materially affect performance. Finally, the current numerical results are development/replay observations. Physical validation is required to determine actual range, accuracy, latency, energy consumption, reliability, and robustness under wind, vibration, multipath, thermal clutter, and obstruction.

IX. FUTURE SCOPE

Future development will focus on physical integration and calibration of the optical, thermal, and acoustic sensors with the flight controller, followed by real-time embedded implementation of signal processing and TDOA/AOA localization. Advanced state-estimation methods such as Kalman filtering, extended Kalman filtering, and nonlinear estimation can be incorporated to improve confidence-aware sensor fusion [25], [50]. AI-based signal classification can also be explored to distinguish beacon signals from environmental interference, while adaptive transition thresholds can be introduced according to signal quality and localization uncertainty.

Further work can improve beacon transmitter and receiver design to extend the effective detection range and support multi-drone or multi-base-station operation. Secure authenticated telemetry and command channels can be expanded to strengthen operational safety. Hardware-in-the-loop testing should precede controlled indoor and outdoor GPS-denied flight experiments so that timing, calibration, estimator, and control failures can be identified before free flight. Finally, onboard computation, payload, and power consumption can be optimized to improve the practicality of the complete autonomous platform [3], [31], [62].

X. CONCLUSION

This paper presented a multi-modal periodic energy-based navigation system for GPS-independent autonomous drone homing. The architecture combines optical, thermal, and acoustic sensing so that each modality contributes at the stage where it is most useful. Optical and thermal sensing support long-range target acquisition, while microphone-array TDOA/AOA processing provides near-field directional refinement. A confidence-aware fusion layer connects these measurements to navigation states, closed-loop control, telemetry, and safety monitoring.

The current development and replay observations show a structured workflow with a 100 m initial homing range, a 20 m near-field transition, 0.8 m final alignment, 10 Hz navigation

updates, 92.4% fused confidence, 1.5 m average localization error, approximately 90% replay homing success, and 10–20 Hz telemetry. These figures are explicitly development-stage observations and require physical flight validation before being treated as general performance claims.

The principal contribution is therefore the system-level staged integration of periodic optical, thermal, and acoustic homing cues into a complete navigation architecture. The next phase is controlled hardware-in-the-loop and flight experimentation under realistic noise, obstruction, lighting, thermal, and communication conditions.

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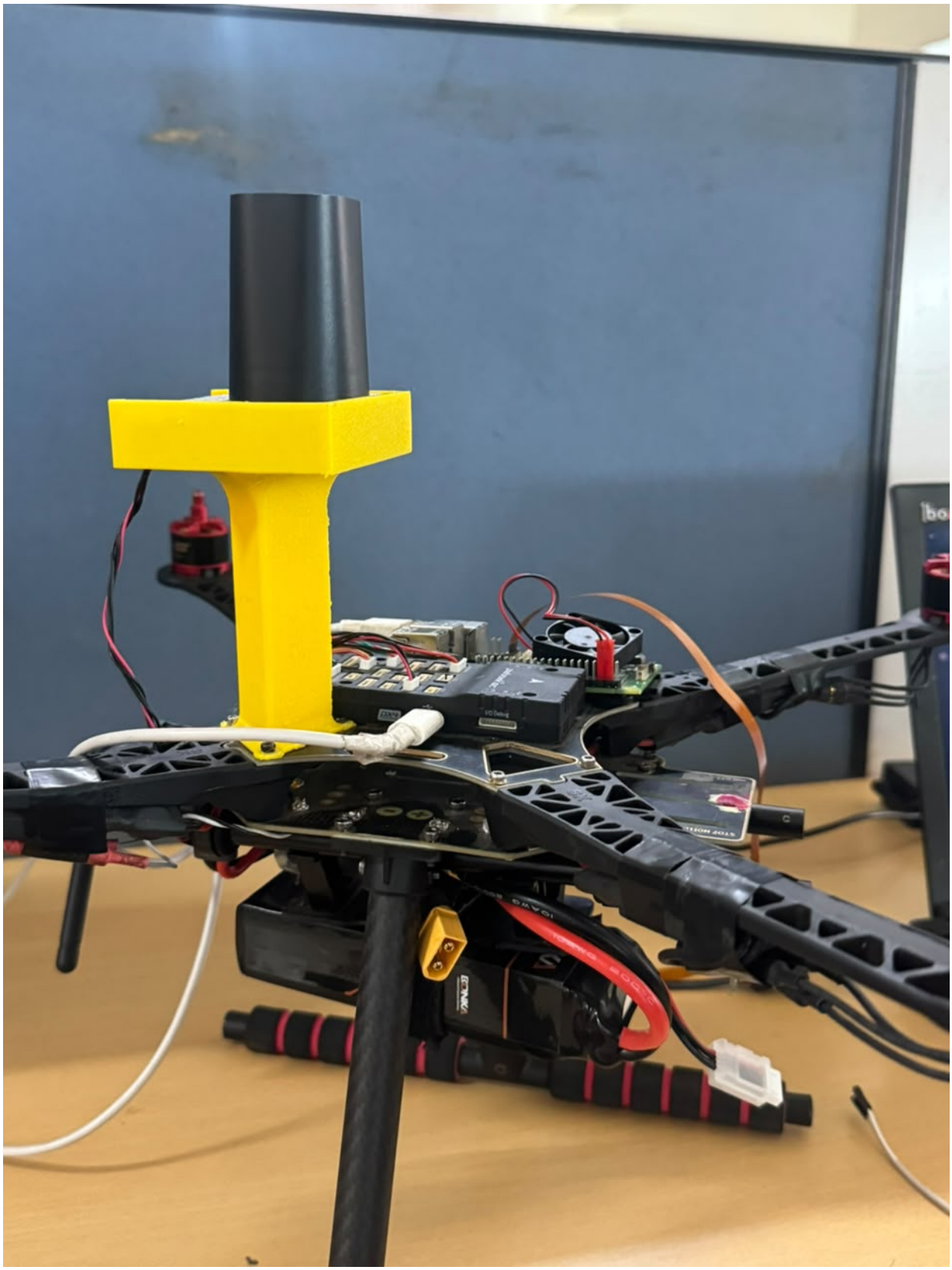


Fig. 6. Actual UAV prototype implementation.

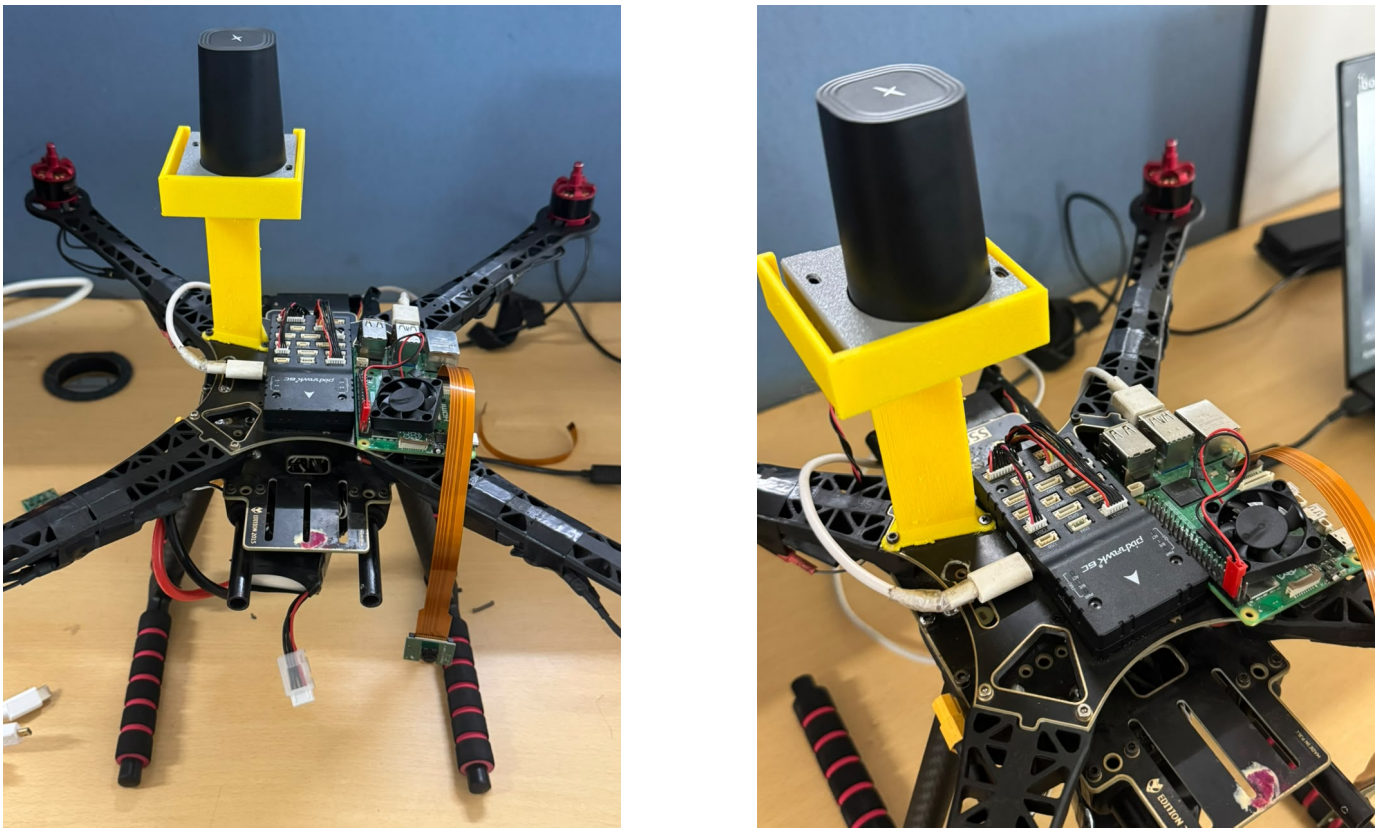


Fig. 7. Additional views of the implemented UAV prototype and sensor assembly.